

PROJECT PROPOSAL · DATA MINING PIPELINE

Predictive Maintenance for Metro Climate & Air Systems

Contextual anomaly detection on 1.5M sensor readings from a metro Air Production Unit

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Dataset: MetroPT-3 (UCI ML Repository) · Porto Metro APU · Feb–Sep 2020

1,516,948

readings

15

sensors

213

days

4

logged failures

Why This Matters

A compressor fault reaches passengers long before it reaches a workshop.

1

Service reliability

An unplanned APU failure strands a train mid-line. Recovery is measured in hours, not minutes, and cascades across the timetable.

2

Passenger comfort

The APU supplies the pneumatic circuit behind climate control, doors and suspension. Air loss degrades cabin conditions before anyone reports a fault.

3

Energy cost

In the four logged failures the compressor ran under load 99.6% of the time versus 14.6% normally, a ~6.8x increase in duty cycle, and a correspondingly large increase in energy drawn.

4

Maintenance economics

Calendar-based servicing either replaces healthy parts or arrives late. Condition-based scheduling targets the intervention window instead.

Problem statement: given only unlabelled multivariate sensor streams, detect degradation of the air production unit early enough and precisely enough to act on.

Related Work and the Gap

Research on this dataset converges from several directions. None of them closes it.

Physics & rule-based HVAC diagnostics

ASHRAE RP-1043 style residual checks and expert rules.

Strong interpretability, but thresholds are tuned per unit and do not transfer across duty cycles.

Deep anomaly detection on APU data

Davari et al. (2021) apply sparse autoencoders and LSTM-AE to this exact unit.

Reconstruction error flags the event, but offers little on why, and evaluation is largely per-sample.

Streaming & explainable PdM

Barros et al. (2020) and Gama et al. (2022) on drift-aware online detection.

Method-focused. Cycle-level structure of the compressor is left on the table.

The gap this project fills

Prior work treats each reading as the unit of analysis. This project treats the compression cycle as the unit of analysis, **then evaluates on an event basis with an explicit lead-time budget.**

The contribution is a reproducible, cycle-aware baseline that reports precision at a fixed alert budget, not per-sample accuracy on an imbalanced stream.

The Data: Four V's

Profiled directly from the source file, not from the dataset card.

VOLUME

1,516,948

readings x 15 sensors = 22.8M values, 208 MB raw CSV covering 213 consecutive days.

VARIETY

7 + 8

Seven analogue channels (bar, °C, A) and eight digital state flags. Mixed types, one clock.

VELOCITY

0.1 Hz

One reading every 10s, 8,640 per day. Batch now, but sized for a streaming deployment.

VERACITY

3.35%

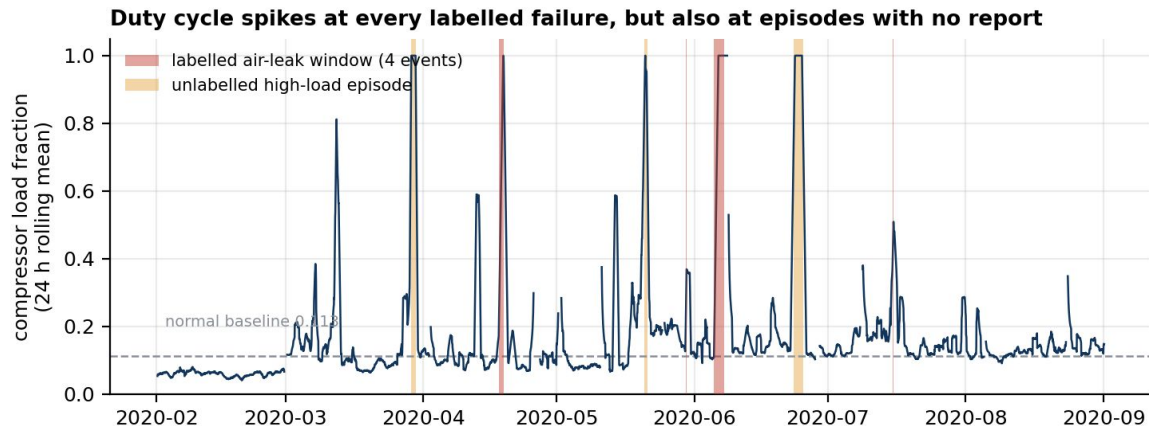
of rows are frozen across all analogue channels; pressures read slightly negative at rest.

Veracity findings that shape the whole pipeline

- TP3 and Reservoirs are perfectly collinear ($r = 1.00$), so one is redundant.
- Only 1.98% of rows fall inside a reported failure window, which is severe class imbalance.
- 331 timing gaps exceed 60s, the largest 48 hours, so the clock is not uniform.
- Labels are event reports, not annotations: unlabelled high-load episodes exist (see next slide).

What Exploration Already Shows

The signal is real. So is the hard part.



Read this way

Every labelled failure sits on a duty-cycle spike. The physics is visible.

But several equally large spikes carry no failure report at all.

A naive threshold detector therefore reaches recall 0.66 at precision 0.21, giving F1 of 0.32.

6.8x

duty cycle inside
failure windows

0.32

F1 of the naive
threshold baseline

75%

of high-load hours
have no report

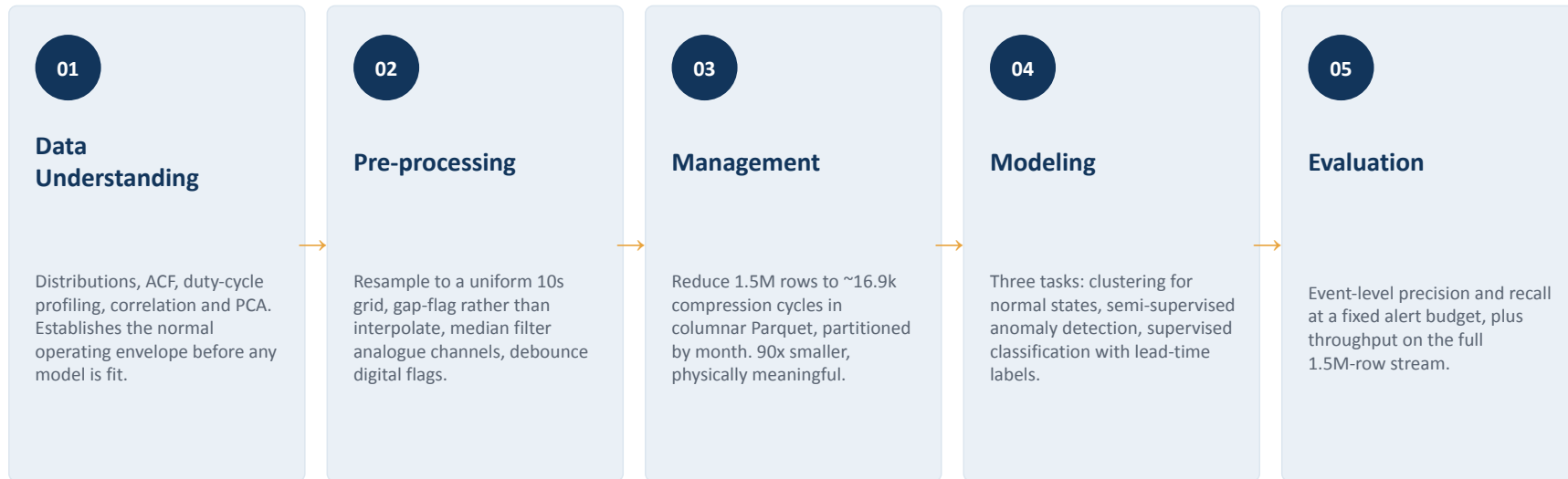
36-62h

precursor lead time
in 2 of 4 events

Implication: the modelling problem is not detection, it is discrimination, separating leak-driven load from demand-driven load.

Methodology: The Whole Pipeline

Five stages, each with a stated decision and a stated reason.



Key design decision: the compression cycle, not the 10-second reading, is the unit of analysis. A leak does not change any single reading much; it changes how long the compressor must run to refill the reservoir. Cycle-level features make that directly measurable.

Three Modeling Tasks, Three Purposes

Chosen because the label situation demands all three, not because more is better.

1

Clustering

K-means and DBSCAN on cycle features

Discovers the operating regimes without labels. Answers: how many distinct 'normal' modes does this unit actually have? Silhouette and Davies-Bouldin select k.

2

Anomaly detection

Isolation Forest and LSTM autoencoder

Trained on February–March only, where no failure is reported. Everything after is scored as deviation. Chosen because labels are scarce and possibly incomplete.

3

Classification

Gradient boosting with lead-time labels

Labels each cycle by hours-to-failure. Answers the operational question: can we call it 24 hours out? Gives calibrated probabilities and SHAP attribution.

Why not go straight to deep learning? Because a tuned Isolation Forest on 40 cycle features that beats $F1 = 0.32$ is a defensible result. An LSTM that beats it without an interpretable baseline to compare against is not.

Evaluation Plan

Accuracy is the wrong metric here. This slide says what replaces it.

Effectiveness

- Precision, recall, F1, PR-AUC. Not accuracy: 'always normal' scores 98.0%.
- Event-level detection: caught if any alarm fires in the window.
- False alarms per 30 days at a fixed alert budget.
- Mean lead time before failure onset, reported with its variance.

Efficiency

- End-to-end wall clock over all 1,516,948 rows, per stage.
- Throughput in readings/second and peak memory, on fixed hardware.
- Per-cycle scoring latency, well under the ~18 min cycle interval.
- Training time and model size, for the edge-deployment question.

Experimental setup

Chronological split, never random. Feb- Mar trains, Apr - Sep tests. Feb- Mar is the longest stretch with no reported failure, which is the whole point: the detector must learn normality from clean data.

Blocked K-fold (k = 5) with a purge gap. Folds are contiguous time blocks separated by a 24h purge window so no cycle straddles a boundary. Standard K-fold is invalid on time series.

Baselines to beat: the naive duty-cycle threshold ($F1 = 0.32$) and a seasonal-naive persistence model. Any proposed method must beat both to justify its complexity.

Timeline

9

21 June to 15 August 2027, with the risky work deliberately placed early.

W1 21-27 Jun

Profiling & EDA

Distributions, ACF, regime characterisation, cycle segmentation validated by hand.

W2 28 Jun-4 Jul

Cleaning & feature store

Uniform grid, gap flags, filtering. Parquet cycle table locked and versioned.

W3 5-11 Jul

Clustering & regimes

K-means / DBSCAN, k selected.
Deliverable: labelled operating-mode map.

W4 12-18 Jul

Anomaly detection

Isolation Forest baseline on clean-window training. First event-level scores.

W5 19-25 Jul

Supervised models

Lead-time labelling, gradient boosting, blocked CV, SHAP attribution.

W6 26 Jul-1 Aug

Efficiency benchmarking

Throughput, latency and memory on the full stream. LSTM autoencoder comparison if schedule allows.

W7 2-8 Aug

Ablations & writing

Ablations, sensitivity to the alert budget, consolidate results, draft report.

W8 9-15 Aug

Buffer & delivery

Reserved for overrun. Final report, deck and reproducible repository.

Weeks 1 and 2 are load-bearing: if cycle segmentation is wrong, every downstream result is wrong. Week 8, the week of 9 August, is unallocated on purpose.

Expected Outcome

What a supervisor will be able to see at the end of week eight.

1 A working detector

Cycle-level anomaly scores for all 213 days, with event-level precision, recall and lead time reported against the four documented failures.

2 A defensible baseline

A published comparison against the naive threshold ($F1 = 0.32$) so any gain is attributable to method, not to metric choice.

3 An explanation, not just a flag

SHAP attribution naming which sensors drive each alarm, which is the difference between an alert and an actionable work order.

4 A reproducible artefact

Versioned Parquet feature store, seeded notebooks and benchmark scripts that rerun end to end on the full 1.5M rows.

If the primary approach fails, the fallback is a change-point detector on the duty-cycle series alone. It is weaker, but it is guaranteed to produce a result.